

THE ADVERSARIAL REASONING METHOD (ARM)

A Conceptual Framework Proposal for Structured Clinical Diagnostic Reasoning

Claudia West | July 2026 | One-Page Summary | Not Yet Empirically Validated

The Problem: Diagnostic error affects an estimated 12 million U.S. outpatients annually (Singh et al., 2014), and autopsy studies show a persistent 10-40% discordance rate between clinical and post-mortem diagnoses (Hoppe et al., 2025). Many of these errors stem not from gaps in medical knowledge, but from failures in the reasoning process itself (premature closure, confirmation bias, anchoring, and the absence of self-critique).

The Proposal: ARM is an eight-step procedural sequence — Observe, Represent, Compete, Attack, Challenge, Update, Reflect, Integrate — that is intended to walk a clinician through a structured hypothesis generation, self-disconfirmation, and reflection post-case as one fully connected process. Each individual step targets a specific, previously documented failure mode in clinical cognition.

The Eight Steps

#	Step	Core Function	Failure Mode Addressed
1	Observe	Structured intake; separation observation from interpretation	Incomplete data gathering
2	Represent	Compression findings into a pattern-oriented summary (Bordage)	Diagnostic ambiguity
3	Compete	Generate competing hypotheses with supporting/contradicting evidence	Premature closure
4	Attack	Mandatory adversarial audit of the leading hypothesis (Croskerry)	Confirmation bias, anchoring
5	Challenge	Select tests by their power to shift the differential, rather than confirm	Low-value testing
6	Update	Iterative revision as new data accumulate	Diagnostic inertia
7	Reflect	Structured post-case self-evaluation	Failure to learn
8	Integrate	Long-term pattern formation across cases	Passive experience accumulation

Honest Positioning: ARM does not claim new cognitive theory. Steps 1, 2, 3, 5, and 6 are procedural restatements of Bordage’s problem representation and Kassirer’s hypothetico-deductive method. The paper’s only structurally original elements are making Croskerry’s disconfirmation strategy a mandatory, fixed position step, and closing the loop with the Reflect and Integrate stages, which have no clear precedent across these three models. Where no defensible claim to novelty exists, the paper states this directly rather than implying otherwise.

Proposed Next Step: A small feasibility pilot (5-8 de-identified case vignettes; 2-4 raters) testing inter-rater agreement on hypothesis identification, documentation time per step, and whether the framework surfaces reconsidered diagnoses that unstructured reasoning would plausibly miss. This is scoped explicitly as feasibility work, not efficacy validation.

Status: Conceptual proposal, not yet empirically tested. No claim of superiority over existing models is made. The full white paper includes literature positioning, pilot design, implementation constraints, and validation roadmap.